

VISHNU KAKARAPARTHI

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Education

Ph.D. in Computer Science
Master of Business Administration
Master of Computer Science
Bachelor of Computer Science and Engineering

Arizona State University
Quantic School of Business and Technology
Arizona State University
SRM University

Work Experience

DealPrint

Jun 2026 – Present

Founding AI and Software Engineer

Remote

- Shipped a production **document classifier** for a 464-type M&A due-diligence taxonomy using Titan embeddings, calibrated confidence, and out-of-taxonomy detection, improving top-1 accuracy from **0.27 to 0.79** while reducing prompt-token usage by **95%** from 52K to ~2K tokens.
- Built DealPrint's internal **agentic customer lifecycle platform** using FastAPI, Next.js, Claude on Bedrock, and Terraform, guiding buyers through 7 gated deal phases and reducing advisor tracking time by **~10x** per phase.
- Delivered **11 human-in-the-loop AI agents** for thesis drafting, qualification interviews, sourcing, diligence, market research, outreach, and delisting monitoring.
- Developed 6 **deal calculators** on a unit-tested finance engine with IRR solvers, SBA 7(a) fees, DSCR stress grids, inverse max-offer solves, and cited assumptions.
- Automated **fact-ledger to deal-memory** updates, increasing populated profile fields by **55%** across 51 deals and improving **94%** of deal profiles.
- Built a **LiveKit-based outbound AI voice agent** on GPT-4.1 for buyer and seller campaigns with FAQ handling, demo booking, voicemail detection, scheduled retries, outcome logging, and a Terraform-deployed Lambda webhook.

Movement Interactive – HijiBand

Jan 2025 – Aug 2025

AI Researcher

Tempe, AZ

- Developing a machine learning-based **fall detection system** using wearable motion and physiological sensors to enhance safety for older adults.
- Specializing in **post-fall analysis** to determine assistance needs, integrating real-time AI motion analysis and biometric monitoring.
- Optimizing models for **high-accuracy detection**, reducing false positives and improving response time through adaptive algorithms.

Boomi Inc

May 2024 – Aug 2024

Gen AI Engineering Intern

Remote

- Developed AI-powered documentation tools using Prompt Engineering, Retrieval-Augmented Generation (RAG), and dynamic prompt classification, **reduced manual effort by 10x and improved accuracy by 95%**.
- Integrated Chainlit to build real-time conversational AI experiences, **increasing user engagement by 70%** and reducing support tickets by 5%.

BrainChip Inc — Zalmotek, Mercedes-Benz

Jun 2023 – Aug 2023

Machine Learning Intern

Laguna Hills, CA

- Led a multi-modal neuromorphic-based anomaly detection initiative, using quantized models on AKD1000 chips, yielded **4x smaller size models, 500x energy savings, and 4x faster inference** over traditional systems.
- Collaborated on distracted driving detection systems with sensor integration, positioning the work for application in the Mercedes-Benz Vision EQXX Concept.

Toy Upgrade (ASU Startup)

Jun 2018 – Jan 2020

Head of Engineering

Tempe, AZ

- Led a 5-member team to create an educational toy with speech interaction, **boosting user engagement by 300%**.
- Enhanced pronunciation matching accuracy by 20% using the SoapBox Labs API, tailored to children's speech patterns.

Heyludwig – Partners Dog Training (Startup)

Jun 2018 – Jan 2020

Technical Lead

Tempe, AZ

- Built and deployed a chatbot using Dialogflow to streamline client interaction, achieving a **25% increase in engagement and 15% reduction in response time**.

Ericsson India Pvt Ltd.

Jan 2016 – Apr 2016

R&D Intern

Gurugram, India

- Led a churn prediction project using Apache Spark and ML libraries, **achieving 64.8% accuracy** by analyzing call detail records.
- Built a multiclass classifier for Facebook stream analytics, **reaching 87.6% accuracy** in categorizing user posts into service-related segments.

Research Experience

Ph.D. Research

Doctoral Researcher

Jul 2019 – Present

Arizona State University

- Led the development of groundbreaking wrist-centric technology and innovative machine learning algorithms for a wrist-worn camera device, advancing human action understanding and expanding its applications.
- Co-PI for the Global Sport Institute and WearTech initiative grants, creating tools for sports enthusiasts and elderly individuals to track physical activities and manage pill-taking effectively.

Master's Research

Master's Researcher

Jan 2018 – Dec 2018

Arizona State University

- Collaborated with global brands like Adidas, Pizza Hut, and Edgenuity to develop tools leveraging affective computing and Tobii eye-tracking technology, revolutionizing user experience data collection and analysis.
- Conducted cutting-edge research on cognitive activity using Brain-Computer Interfaces (BCI), EEG, GSR data, and eye-tracking to model trust and motivation, optimizing e-commerce sales through A/B testing.

SRM University

Bachelor's Thesis Research

Jan 2016 – Jul 2016

Kattankulathur, India

- Evaluated diverse machine learning methods to predict Indian stock market closing prices, achieving 0.79 accuracy for Random Forests and 0.77 for Deep Neural Networks.

Projects

ORBIT – Open Research-driven Backtesting Intelligence Tool

Jan 2026 – Present

- Designed and orchestrated an autonomous multi-agent research system that decomposes natural-language trading hypotheses into structured experiments, dynamically generates strategies, executes institutional-grade backtests, and iteratively refines results through agent-driven critique and feedback loops.
- Implemented role-specialized agents (Hypothesis Analyst, Strategy Builder, Backtest Executor, Risk Auditor, and Report Synthesizer) coordinated via Model Context Protocol (MCP), enabling reproducible experiment lineage tracking, tool-augmented reasoning, and modular deployment for transparent and scalable quantitative research.

AutoTrader Bot: Automated Stock Trading Platform

Aug 2025 – Dec 2025

- Built an end-to-end trading automation system using **n8n** and the **Charles Schwab API**, integrating real-time market data, financial news feeds, and semantic analysis for informed portfolio decisions and autonomous trade execution with stop-loss and profit-taking logic.
- Engineered modular workflows for dynamic asset allocation across mid-, low-, and high-cap stocks, combining quantitative signals with news-driven sentiment to deliver real-time performance alerts and support AI-driven predictive modeling.

AutoResumeGen: Multi-Agent Resume Generator

Jun 2025 – Dec 2025

- Built a modular system using Crew AI and GPT-4o to generate tailored, ATS-optimized resumes from job descriptions and user profiles.
- Designed and deployed agents including **RoleParser**, **PersonaCalibrator**, **BulletRewriter**, and **PDFExporter** with asynchronous orchestration via LangChain and RAG-enabled vector search.
- Integrated input preprocessing with **JDParser** and dynamic persona modeling, and implemented a **ResumeCritiqueAgent** for LLM-based feedback on tone, clarity, and keyword alignment.
- Integrated a **MockInterviewAgent** to generate contextual interview questions and model responses tailored to the resume and job description, supporting candidate preparation and reflection.

Dementia Framework

Jan 2022 – Feb 2022

- Developed and presented a Participatory Design Framework for dementia care technology, integrating policy informatics principles across technology, law, architecture, markets, norms, and education.

Technological Tools for Dementia Care

Aug 2021 – Dec 2021

- Analyzed relationships between assistive technologies and dementia patients; proposed low-fidelity solutions like environmental sensors, addressing usability and location tracking limitations.

LIDAR Object Detection

Feb 2019 – May 2019

- Implemented advanced 3D object detection models (PointNet, PointNet++, VoxelNet) using Python and TensorFlow to enhance robotic perception from LIDAR data via ROS and Open3D.

Vision-Based Manipulator Control with Fetch

Feb 2019 – May 2019

- Applied visual servoing and depth estimation with ROS and OpenCV to guide a Fetch robot in reaching partially occluded objects, addressing challenges like cable entanglement and occlusion.

Bacteria Detection using Transfer Learning

Jun 2018 – Jul 2018

- Developed a production-grade image classification system for NanoCheQ using AWS SageMaker and OpenCV, improving accuracy by 35% over baseline and achieving 94.9% overall accuracy.

Auto-Scaling Image Recognition API	Feb 2018 – May 2018
Built a scalable image classification service with Python and TensorFlow (Inception-V3), leveraging AWS EC2, S3, and SQS to dynamically scale compute for real-time image inference.	
DropBy – Event Crowdsourcing App	Mar 2018 – Apr 2018
Developed a sentiment-aware event app using Java and Android Studio on Google Cloud Platform with features including real-time messaging, auto-scaling, and media sharing.	
Semantic Search on Movie Summaries	Jan 2018 – May 2018
Led a team of four to build a QA system using convolutional neural networks and NLP techniques, including coreference resolution, sentence embedding, and named entity recognition, achieving 54.7% accuracy.	
Spatial Hot Spot Analysis from Geospatial-Temporal Data	Aug 2017 – Nov 2017
Built a spatial analytics pipeline using Apache Spark and Sedona to analyze over 50 GB of NYC Taxi data, identifying high-activity zones via Getis-Ord statistics and real-time visualization.	
Classification of Higgs Boson Particle	Jan 2017 – May 2017
Trained an ensemble of neural networks and random forest classifiers on 11M CERN records using Keras, TensorFlow, and scikit-learn, achieving 71% accuracy in detecting the Higgs Boson.	
Knowledge-Based Question Answering System	Jan 2017 – May 2017
Built a QA system querying DBpedia using Python, Stanford NLP, and SPARQL, with modules for semantic parsing, query formulation, and graph matching focused on firm-related questions.	
Maximizing NYC Taxi Revenue Visualization	Jan 2017 – May 2017
Created a JavaScript-based interactive dashboard using D3, Crossfilter, and dc.js, analyzing 1B+ taxi trips to provide actionable insights for fare, tip, location, and passenger behavior.	
SRM - PURA (Providing Urban Amenities in Rural Areas)	Jan 2013 – Dec 2015
Developed backend systems and multilingual front-end features using PHP and MySQL; enabled e-learning, e-governance, and IT access in underserved rural communities.	
SRM-SE (SRM Search Engine)	Jan 2013 – Dec 2015
Supported server and network administration for a university search engine project that filtered, clustered, and displayed results with improved usability and relevance.	
Publications and Patents	
Striking the Privacy-Model Training Balance: A Case Study using PERACTIV Device, HCII 2024. Link Innovating Medication Adherence for Smart Cities: Leveraging PERACTIV and Automated Annotation Pipeline, International Conference on Smart Multimedia 2024. A Hand-Directed System for Identifying Activities, U.S. Patent App. #20230324993. Link Wrist View: Understanding Human Activity Through Hand, HCII 2023. Lecture Notes in Computer Science, vol 14021. Link PERACTIV: Personalized Activity Monitoring - Ask My Hands, HCII 2022. Link Machine Learning Algorithm Hypothesis on Smart Gyroscopic Tuned Dampers for Earthquake Resistance Building, International Journal of Multidisciplinary Research and Development, vol. 2, pp. 705-707, 2015. Link	
Awards and Honors	
Graduate Mentorship Award, ASU	Jan 2025 – May 2025
Recognized for exceptional mentorship of student researchers and capstone teams across interdisciplinary labs.	
Graduate Teaching Excellence Award, ASU	Aug 2023 – Dec 2023
Awarded for outstanding instructional contributions as an Instructor for FSE 100: Introduction to Engineering.	
WearTech Grant, Greater Phoenix Economic Council (GPEC)	Jul 2021 – Jun 2022
Secured a \$20,000 grant as Co-PI through CUBiC Lab (ASU) to advance wrist-based video analytics for medication adherence in elderly individuals.	
Global Sport Institute Grant, ASU	Jan 2021 – Dec 2021
Awarded \$20,000 as Co-PI to develop wearable tools for tracking physical activity and enhancing remote training for seniors and athletes.	
Third Prize – All India Software Development SESCON-15	2015
Won third place at a national-level software development competition held at Sri Eshwar College of Engineering.	
First Prize – Denken Fest Coding Competition	2014
Won first place in a competitive coding challenge at Aaruush, SRM University.	
Academic Excellence Award – Indus Foundation	2014
Received academic merit recognition for top performance among peers.	
First Prize – CTF (Capture the Flag) at eHack	2013
Won first place in team CTF and placed sixth individually in the cybersecurity challenge hosted by Infysec.	

Teaching Experience

Graduate Teaching Assistant at *School of Computing and Augmented Intelligence (SCAI), ASU* **Jan 2019 – Present**

- Instructor for FSE 100: Introduction to Engineering (Fall 2020; Fall 2022; Fall 2023; Fall 2025); covered engineering design, modeling, communication, and teamwork.
- Taught over **200 students** in CSE 463: Human-Computer Interaction under Dr. Hasti Seifi (Spring 2023); designed course structure, quizzes, and examinations.
- Taught **over 1400 students** in CSE 463 under Dr. Robert Atkinson (Spring 2019, Fall 2019, Spring 2020); supported course design, assessment, and grading.
- Lab Instructor for CSE 110: Principles of Programming with Java (Summer 2019, 2020, 2022); taught Java fundamentals and provided one-on-one mentorship.

Graduate Services Assistant – Grader / Lab Instructor at *SCAI, ASU* **Aug 2017 – Jul 2022**

- Assisted with course materials and grading for CSE 360: Introduction to Software Engineering and CSE 110: Java Programming.
- Taught computer fundamentals to 150 students in CSE 180: Computer Literacy, including Excel, Word, HTML, SQL, Networking, and Cybersecurity.
- Supported instructional design and grading for CSE 463: Human-Computer Interaction.

Capstone Team Mentor **Aug 2017 – Dec 2022**

- **CUBiC Lab (2021–2022)**: Mentored four undergraduates and one volunteer to develop wearable tech using off-the-shelf components and mobile deep learning.
- **iLUX Lab (2017–2020)**: Guided four major projects with over 20 students; contributions included UX feature development for Pizza Hut (Yum! Brands), data pipeline design from EEG, BCI, GSR, AFFDEX, Eye-tracking, and Android app benchmarking with WEKA.
- **ANGLE Lab (2019)**: Mentored two teams to develop AR-based pallet packing optimization systems for FedEx using spatial layout algorithms.

AI Instructor at *AI4ALL* **Jun 2020**

- Designed and taught an AI curriculum to 24 high school students, covering Clustering, Classification, Regression, Naive Bayes, Neural Networks, and Bias through hands-on projects.

Professional Services

Editorial Board Member at *Clareus Scientific Science and Engineering (ISSN: 3065-1182)* **2025 – Present**

Editorial Board Member at *PriMera Scientific Engineering (PSEN)* **2025 – Present**

Readers Committee Member at *International Journal of Artificial Intelligence and Knowledge Engineering (IJAIKE)* **2025 – Present**

Special Sessions Co-Chair at *International Conference on Smart Multimedia (ICSM 2025)* **2025**

Committee Member & Co-Chair at *ACM Symposium on Eye Tracking Research & Applications (ETRA 2025)* **2025**

Reviewer & Program Committee Member for Peer-Reviewed Conferences and Journals **2019 – Present**

- Reviewed over **172** papers, with **50** reviews for top-tier conferences and journals in HCI, computer vision, and multimedia computing.
- Served as a PC member or reviewer for:
 - ACM CHI – Conference on Human Factors in Computing Systems
 - HRI – Conference on Human-Robot Interaction
 - ACM ETRA – Eye Tracking Research and Applications
 - ACM TOMM – Transactions on Multimedia Computing, Communications, and Applications
 - OzCHI – Computer-Human Interaction Conference (Australia)
 - ICWSM – International AAAI Conference on Web and Social Media
 - PacificVis – IEEE Pacific Visualization Symposium
 - HCII – International Conference on Human-Computer Interaction

Administrative Experience

Administrative Researcher at *CUBiC Lab, ASU*

Jan 2019 – Jul 2020

- Secured a **\$20,000 WearTech grant** as Co-PI through the **Center for Cognitive Ubiquitous Computing (CUBiC)** to research medication adherence among elderly individuals living alone, developing novel video-based activity recognition algorithms using a wrist-mounted camera system.
- Awarded a **\$20,000 Global Sport Institute grant** as Co-PI to create computer vision-based tools for tracking physical activity and enhancing remote training experiences for both sports enthusiasts and aging populations.
- Supervised thesis students, capstone teams, and research volunteers in computer vision and human-computer interaction, supporting skill development and contributing to the lab's research pipeline.

Administrative Researcher at *iLUX Lab, ASU*

Jan 2019 – Jul 2020

- Orchestrated daily operations at the **Innovative Learner and User eXperience (iLUX) Lab**, mentoring students and supporting a dynamic, interdisciplinary research team.
- Led projects in Affective Computing and eye-tracking, fostering collaborations and contributing to successful grant proposals.
- Worked with Edgenuity to improve its educational technology platform and partnered with Adidas and Pizza Hut to enhance sales and user experience through data-driven A/B testing.
- Mentored capstone teams and volunteers, advancing research in emotion-aware systems, learning analytics, and user-centered design.

Administrative Researcher at *ANGLE Lab, ASU*

Jan 2019 – Jul 2020

- Led research initiatives in Augmented Reality at the **Advanced Next Generation Learning Environments (ANGLE) Lab**, mentoring graduate and undergraduate researchers.
- Secured research funding through successful grant proposals and coordinated cross-disciplinary collaboration within the lab.
- Partnered with FedEx to develop an Augmented Reality Unit Load Device (ULD) optimization tool using Microsoft HoloLens, improving cargo loading efficiency and spatial utilization.
- Supervised capstone projects and mentored student volunteers in AR/VR development, contributing to hands-on learning and lab research outcomes.

Guest & Invited Talks

Nursing Asia-Pacific 2026

2026

- Presented as a Distinguished Speaker on deploying PERACTIV in nursing homes through centralized monitoring dashboards, AI-driven medication adherence tracking, ADL recognition, and real-time patient analytics to support clinical decision-making, improve patient safety, and reduce caregiver workload.

IEEE Engineering in Medicine and Biology Society (EMBS) Phoenix Chapter

2026

- Delivered an invited EMBS seminar showcasing PERACTIV, a biomedical AI platform combining wearable sensors, interpretable machine learning, and human-centered design for continuous patient monitoring and precision digital health.

IEEE New Era AI 2025 World Leaders Summit

Dec 2025

- Delivered an invited talk at the IEEE New Era AI 2025 World Leaders Summit on PERACTIV, a wrist-centric multimodal AI framework for personalized medication adherence, human activity recognition, and human-centered applications in healthcare, manufacturing, and robotics.

ASU New College B²C² Data and Methods Student Workshop

Nov 2020

- Delivered invited instruction on AI, Python, machine learning, and data analytics to high school and undergraduate students.

Hackathon Projects

WizardEyes

Sep 2021

- Developed a real-time computer vision module for Smart City infrastructure using edge AI, delivering actionable insights such as occupancy, queue length, mask compliance, and crowd density through a unified API.
- Leveraged Luxonis Oak-D-IoT-40 and cloud integration to enable low-latency inference and scalable deployment in urban monitoring systems.

Open-Source Contributions

Mayhem Heroes

Apr 2022 – Jun 2022

- Ranked **#2 globally** in the Mayhem Heroes program for automating fuzz testing across **48 critical open-source repositories**, identifying zero-day vulnerabilities and enhancing software resilience.
- Led containerized deployments and CI/CD pipelines to integrate Mayhem's autonomous security engine into diverse OSS ecosystems, accelerating vulnerability discovery and responsible disclosure workflows.

Technical Skills

Languages: Bash, HTML/CSS, JavaScript, Python, SQL

Developer Tools: Git, Hugging Face Hub, Jupyter, LangChain, MLflow, n8n, REST APIs, Streamlit, VS Code, Weights & Biases

Technologies/Frameworks: Claude (Anthropic), FastAPI, Flask, Gradio, LLaMA, Mistral, OpenAI API, PyTorch, Retrieval-Augmented Generation (RAG), TensorFlow, Transformers (Hugging Face), Vector DBs (ChromaDB, Pinecone, Weaviate)

Cloud/Platforms: AWS (Bedrock, EC2, Lambda, S3, SageMaker), Azure OpenAI, Docker, GCP (Vertex AI), Kubernetes, MLOps, DevOps

Data/Visualization: Elasticsearch, Matplotlib, NumPy, pandas, scikit-learn, seaborn, Tableau